Sidelobe Suppression for High-Resolution SAR Imagery Based on Spectral Reshaping and Feature Statistical Difference

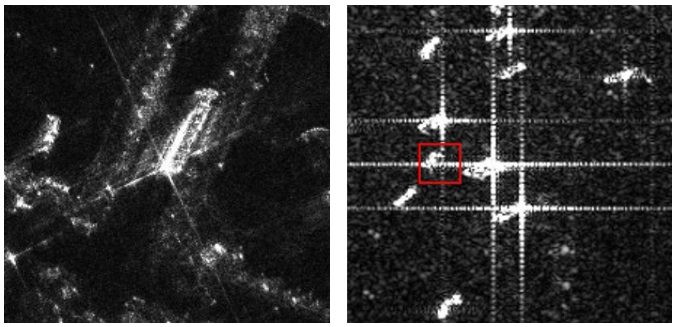
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***Abstract*— Sidelobe suppression is a crucial preprocessing technique for synthetic aperture radar (SAR) image analysis. The presence of strong scattering targets generates sidelobes that can interfere the targets with relatively weak scattering. The overlapping of multiple strong cross-shaped sidelobes may even generate fake targets, significantly influencing the accuracy of SAR target detection and recognition. Among existing sidelobe suppression methods, the spectral reshaping sidelobe reduction (SRSR) method has shown promising results. It separates the mainlobe and sidelobes by altering the sidelobe direction while preserving the SAR image resolution. However, this method exhibits limitations in effectively suppressing strong cross-shaped sidelobes. It also introduces additional sidelobes, blurring the surroundings of the scattering points. This article proposes an improved SRSR method to resolve this disadvantage. It constructs a feature image representing the superposition of sidelobes. This is achieved by analyzing the statistical differences of complex data between orthogonal and nonorthogonal sidelobe regions before and after spectral reshaping. Further modulus selection ensures that the feature image only contains the sidelobe information that needs to be eliminated. The proposed method successfully resolves the drawback of introducing new sidelobes in the original SRSR while achieving better suppression of strong cross-shaped sidelobes. Experimental results on airborne and spaceborne SAR images demonstrate that the proposed method outperforms other state-of-the-art techniques. Improved peak sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR) in both range and azimuth directions and smaller image entropy can be achieved by our method. Due to its superior sidelobe suppression capability, the SAR images processed by our method exhibit significantly improved accuracy in target detection.**

***Index Terms*— Sidelobe suppression, spectral reshaping, synthetic aperture radar (SAR), target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is a high-resolution radar system that has the capability of all-weather and all-day imaging [1], which has been widely applied in industries, such as mapping, urban planning, target detection, and recognition [2]. Due to the limited frequency band and azimuth measurement angle of the radar system, the obtained 2-D spatial spectrum support area of the target has a limited distribution. After undergoing a 2-D Fourier transform, the point spread function (PSF) of the imaging system takes the form of a Sinc function in both azimuth and range directions [3], [4], [5], [6]. The Sinc function exhibits a gradual decrease as the distance from the origin increases. Consequently, a bright scatterer has the ability to distribute its energy to regions of the image that are located some distance away. As shown in Fig. 1, the sidelobe energy severely affects the quality of SAR images. High sidelobes not only obscure weak scattering targets in proximity but also cause overlapped sidelobes, which may create fake targets and influence the accuracy of target recognition and detection [7].



(a) (b)

Fig. 1. (a) Strong cross-shaped sidelobe. (b) Sidelobes obscuring weak scattering targets.

The traditional technique for sidelobe suppression is the linear method, also known as the window filtering method [8]. It smooths and suppresses the sidelobes by applying window filtering functions, such as Hamming, Hanning, and Kaiser in the frequency domain. However, it is difficult to select the most suitable window for sidelobe suppression in SAR image scenes due to different adaptabilities of window functions [9]. To address this drawback, scholars have introduced the window function weighting technique. This technique utilizes weighted networks with transmission functions, such as Hamming window, Kaiser window, Blackman window, and Taylor window for sidelobe suppression [10]. The window function weighting method includes unidirectional weighting and bidirectional weighting. The unidirectional weighting method suppresses sidelobes by weighting the temporal window function of the matched filter coefficient at the receiving end. On the other hand, the bidirectional weighting method further suppresses sidelobes by applying temporal window

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function weighting to the transmitted signal based on the unidirectional weighting method [11]. Nevertheless, the window function weighting method still results in varying degrees of mainlobe broadening at the expense of reducing the peak sidelobes. Simulation results show that the bidirectional weighting method has lower sidelobes. However, it also leads to a more severe increase in mainlobe broadening, resulting in a significant decrease in radar range resolution [12]. Therefore, the linear windowing method is not suitable for processing high-resolution SAR images.

Another technique is spatial variant apodization (SVA) [13]. SVA is a nonlinear weighting algorithm that directly suppresses sidelobes in the image domain. SVA uses a three-point convolution to weight each pixel with a cosine-on-pedestal function and finally selects the minimum value of multiple sample points as the output value. However, SVA can only be used for images under integer Nyquist sampling. To address this drawback, Smith [14] conducted research based on SVA and proposed general SVA (GSVA). Experimental results on SAR data of the Michigan Stadium showed that GSVA can handle images under different Nyquist sampling rates, but the sidelobe suppression performance is unstable and less robust under different sampling rates [15]. Castillo-Rubio et al. [16] proposed robust SVA (RSVA) based on GSVA, which replaces the three-point convolution in SVA with a five-point convolution. It also suppresses sidelobes in cases of noninteger Nyquist sampling. Simulation results showed that RSVA has better sidelobe suppression performance than GSVA, but it may cause energy loss in the mainlobe. To solve the energy loss problem of RSVA, Yuan et al. [17] combined a convolutional neural network (CNN) to compensate the energy of the mainlobe region. Experimental results on TerraSAR-X data showed that this method performed better in terms of sidelobe suppression and signal-to-noise ratio maintaining. Liu et al. [18] used a bidirectional recurrent neural network (BiRNN) to extract the mainlobe and mainlobe energy, achieving sidelobe suppression at any Nyquist sampling rate. Ni et al. [19] improved RSVA and further proposed modified SVA (MSVA), which expanded the traditional filter from a three-point to a five-point filter. Based on different sampling rates, corresponding filter parameters were set to obtain the optimal solution satisfying constrained optimization theory, enabling the processing of images at any Nyquist sampling rate. As a result, MSVA achieved good sidelobe suppression results on RADARSAT and airborne SAR data. Xiong et al. [20] improved the sidelobe suppression performance of SVA by adjusting the upper and lower limits of the cosine basis weighting function. Experimental results on the X-band SAR images showed improvements in sidelobe suppression and mainlobe sharpening. However, since the SVA series algorithms are based on pixel-by-pixel processing techniques, the computational speed is significantly reduced when dealing with large amounts of data.

Introducing certain algorithms during the imaging stage can also achieve sidelobe suppression. Shengwei et al. [21] proposed an image self-focusing algorithm based on improved spectral compression, which achieves self-focusing and azimuth sidelobe suppression through compressing the full aperture image spectrum. However, this method cannot be applied to imaging modes such as stripmap that do not involve beam rotation and is deeply tied to the back-projection (BP) imaging mode. Yuan [22] proposed a sidelobe suppression method based on spatially smoothed coherence factors. Sidelobe suppression is achieved by calculating the coherence between echo signals using partially overlapped subarrays of equal length. However, the results show that this method leads to significant energy loss of the land areas and is only suitable for handling strong scattering point targets. In fact, sidelobe suppression during the imaging stage not only consumes more computational resources but may also result in information loss of raw data. Therefore, in practical applications, it is more common to perform sidelobe suppression after obtaining the raw data.

During the study of the ultrawideband (UWB) SAR spectrum support area, it was found that the sidelobe direction is closely related to the shape of the spectrum support area [23], [24], [25]. Changes in the shape of the spectrum support area will lead to changes in the sidelobe direction. Wang et al. [26] found that changing the shape of the spectrum support area would change the sidelobe direction, whereas the mainlobe width and amplitude remain almost unchanged. When two images with the same mainlobe position and amplitude but different sidelobe directions are obtained, simple mathematical logic operations can be used to separate the mainlobe and sidelobes, thereby suppressing the sidelobes. Wang et al. [26] named this technique the spectral reshaping sidelobe reduction (SRSR) method. Simulation data and outdoor aperture radar experiments showed that the SRSR method effectively suppresses most sidelobes while maintaining the SAR image resolution. However, this method can generate new sidelobes and make the surroundings of the scattering points blurry. To address the drawback of new sidelobe generation, An [27] proposed an improved SRSR method based on spectral weighting. After changing the spectrum support area, different weighting values are assigned to different frequency spectrum regions to achieve better suppression result. The results proved that An's method has a superior suppression performance on the first sidelobe compared with the original SRSR method. However, sidelobe suppression methods based on spectral reshaping theory still have problems of blurred surroundings of scattering points and poor suppression performance on strong cross-shaped sidelobes.

This article proposes an improved SRSR algorithm. It utilizes the statistical differences in complex data between the mainlobe regions and orthogonal/nonorthogonal sidelobe regions of scattering points before and after spectral reshaping. By applying mathematical logic operations, a feature image with superimposed scattering is constructed. To avoid energy loss at strong scattering points, modulus selection is applied to the feature image before performing logical operations. The experimental results using airborne and spaceborne SAR images demonstrate the effectiveness of the proposed method in eliminating the new sidelobes generated by the original SRSR method. More importantly, there is no drawback that blurring the surroundings of scattering points. Moreover, it

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significantly enhances the suppression performance on strong cross-shaped sidelobes.

The rest of this article is organized as follows. Section II introduces the principles of the original SRSR method and discusses the challenges it faces. Section III presents the proposed sidelobe suppression method based on SRSR. Section IV presents the sidelobe suppression results of SAR images with different frequency bands by comparing the proposed method with MSVA, RSVA, and original SRSR method. The comparative experiments on target detection were also conducted in Section IV to demonstrate the practicality of the proposed method. Section V presents our conclusion.

II. Original SRSR Method

A. Principle of Spectral Reshaping

Fourier imaging technology [28] is commonly used to solve electromagnetic inverse scattering problems in the radar field, mainly using 2-D Fourier transform for spectral domain imaging [29]. The PSF is one of the basic indicators for measuring the imaging quality of radar, and it is also an important parameter for describing the system properties. Due to the limited bandwidth of the transmitting signal and system hardware, the position and shape of ideal point targets cannot be fully reconstructed from the echo data. As a result, the reconstructed points in actual imaging are extended with a certain degree of blurring. The 2-D spectrum of the imaging data obtained by the SAR system is often referred to as the spectral support area. It is typically a continuous distribution represented by a 2-D function, which undergoes mutual transformation through the 2-D Fourier transform/inverse transform with the PSF.

The 1-D Sinc function is the inverse Fourier transform of a rectangular window with a width of 2π and height of 1, as described by the following formula:

$$\sin c(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \exp\{j\omega t\}d\omega = \frac{\sin(\pi t)}{\pi t}. \tag{1}$$

By further derivation, it can be obtained that the 2-D Sinc function is the 2-D inverse Fourier transform of a rectangular window with the range of $2\pi \times 2\pi$ and height of 1, as described by the following formula:

$$\begin{aligned}\sin c(t,u) &= \frac{1}{2\pi}\frac{1}{2\pi}\int_{-\pi}^{\pi}\int_{-\pi}^{\pi} \exp\{j(\Omega u + \omega t)\}d\omega d\Omega \\ &= \frac{\sin(\pi t)}{\pi t}\cdot\frac{\sin(\pi u)}{\pi u}.\end{aligned} \tag{2}$$

The 2-D Sinc function can be regarded as the product of two 1-D Sinc functions, while the PSF of the rectangular spectral support area is a 2-D Sinc function. It manifests as "cross-shaped" sidelobes in the SAR image. The spectral support area and the PSF possess the properties of 2-D Fourier transform, namely, rotation invariance and distortion characteristics. Rotation invariance means that when one domain rotates by an angle of θ, and the other domain also rotates by the same angle of θ. Distortion characteristics mean that if one domain is distorted along one data axis, the other domain will also be distorted along the other axis with equal magnitude.

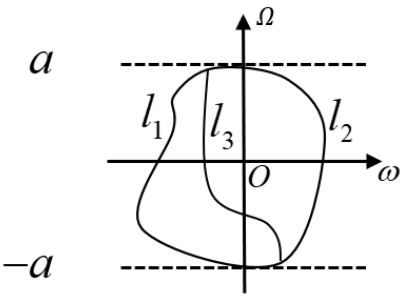


Fig. 2. Relationship between the Fourier transform and boundary.

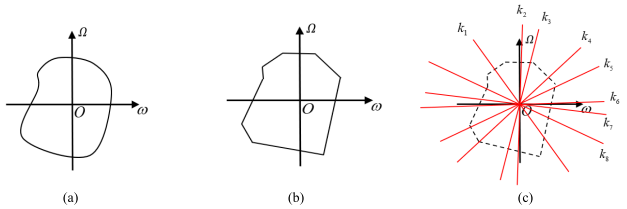


Fig. 3. Quantity and direction of sidelobes of any uniform plane. (a) Any uniform plane. (b) Approximate polygon. (c) Corresponding sidelobe directions.

The spectral support area obtained by radar systems is a 2-D continuous plane, which can be seen as the combination of many small rectangular planes. According to the properties of the 2-D Fourier transform, it can be deduced that the 2-D inverse Fourier transform of a uniformly continuous plane enclosed by an arbitrary 2-D closed curve is a complex weighted sum. This sum consists of a group of 2-D Sinc functions, with their peak vertices located at the origin. Fig. 2 shows a uniformly continuous plane enclosed by an irregular 2-D closed curve.

According to the Fourier transform, the boundary relationship is depicted as follows:

$$F_1(t,u) = \frac{1}{2\pi}\frac{1}{2\pi}\int_{-a}^{a}\int_{l_1}^{l_3} \exp\{j(\omega t + \Omega u)\}d\omega d\Omega \tag{3}$$

$$F_2(t,u) = \frac{1}{2\pi}\frac{1}{2\pi}\int_{-a}^{a}\int_{l_3}^{l_2} \exp\{j(\omega t + \Omega u)\}d\omega d\Omega \tag{4}$$

$$\begin{aligned}F(t,u) &= F_1(t,u) + F_2(t,u) \\ &= \frac{1}{2\pi}\frac{1}{2\pi}\int_{-a}^{a}\int_{l_1}^{l_2} \exp\{j(\omega t + \Omega u)\}d\omega d\Omega.\end{aligned} \tag{5}$$

It can be deduced that the 2-D inverse Fourier transform of a continuous plane is solely determined by its enclosed outer boundary. Any uniform continuous plane can be approximated by a polygon. Suppose that this polygon has m edges, then the plane has m distinct derivative values. The absolute value of the corresponding PSF of this plane has a total of m sidelobes. All these sidelobe directions pass through the origin as straight lines, with their slopes precisely being the negative reciprocals of the m distinct derivatives [30], [31], as shown in Fig. 3.

The PSF corresponding to the ideal rectangular spatial spectrum is cross-shaped, without nonorthogonal sidelobes. When changing the shape of the spectral support area, different nonorthogonal sidelobes will appear. As shown in Fig. 4, assuming the correspondence between the support area and

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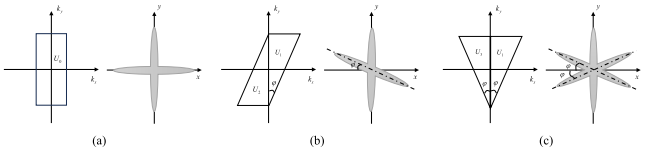


Fig. 4. PSF corresponds to different shapes of spectral support areas. (a) Rectangular. (b) Parallelogram. (c) Triangle.

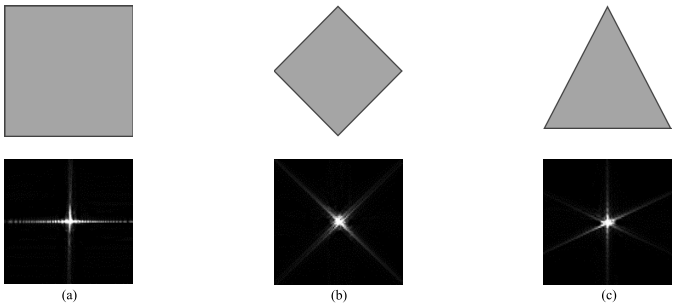


Fig. 5. PSF corresponds to different shapes of spectral support areas. (a) Rectangular. (b) Rhombus. (c) Triangle.

the PSF is denoted by $U_0(k_x, k_y) \leftrightarrow u_0(x, y)$, $U_1(k_x, k_y) \leftrightarrow u_1(x, y)$, $U_2(k_x, k_y) \leftrightarrow u_2(x, y)$, $U_3(k_x, k_y) \leftrightarrow u_3(x, y)$, the support area $(U_1 + U_2)$ composed of U_1 and U_2 can be considered as a distorted deformation of the rectangular support area U_0.

According to the distortion characteristics of Fourier transform, the relationship between U_0 and u_0 is

$$U_0\big(k_x + k_y \tan\varphi, k_y\big) \leftrightarrow u_0(x, y - x\tan\varphi). \tag{6}$$

Then, the azimuth sidelobe of the PSF changes from being along the x-axis to a direction at an angle φ with the x-axis, as shown in Fig. 4(b). According to the rotation invariance of the Fourier transform, it can be deduced that

$$U_1\big(k_x, k_y\big) = U_2\big(-k_x, -k_y\big) \leftrightarrow u_1(x, y) = u_2(-x, -y). \tag{7}$$

In other words, $u_2(x, y)$ is the result of rotating $u_1(x, y)$ by 180°. As $u(x, y) = u_1(x, y) + u_2(x, y)$, the sidelobe directions of $u(x, y)$, $u_1(x, y)$, and $u_2(x, y)$ are the same. Through the above derivation process, the corresponding relationship shown in Fig. 4(c) can also be obtained.

Fig. 5 shows the real PSF corresponding to rectangular, rhombus, and triangular spectral support areas, respectively.

B. Description of the Original SRSR Method

The original SRSR method alters the shape of the spectral support area by cutting [32], resulting in a new image with different sidelobe directions from the original. Fig. 6 displays the flowchart of the original SRSR method.

The input SAR data are complex, and the spatial–spectral support area is obtained through 2-D Fourier transform. To facilitate analysis, we define $F(k_x, k_y)$ as the distribution function of the rectangular spectral support area of the original data and $F'(k_x, k_y)$ as the distribution function of the spectral support area after spectral reshaping. The PSF before and after

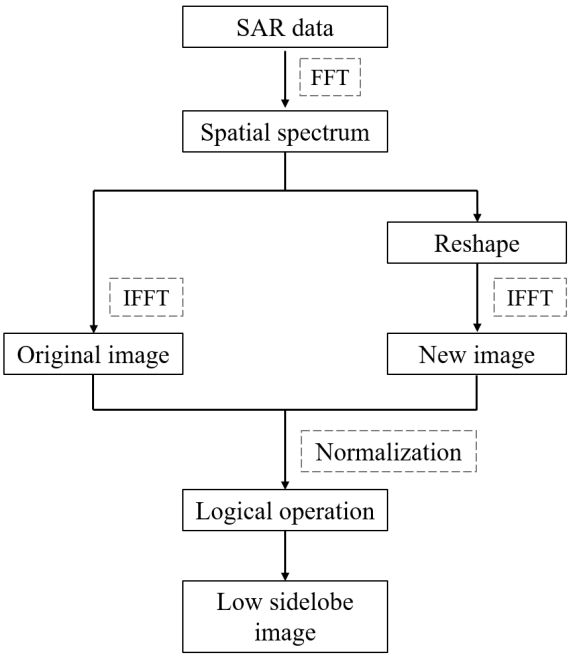


Fig. 6. Flowchart of the original SRSR method.

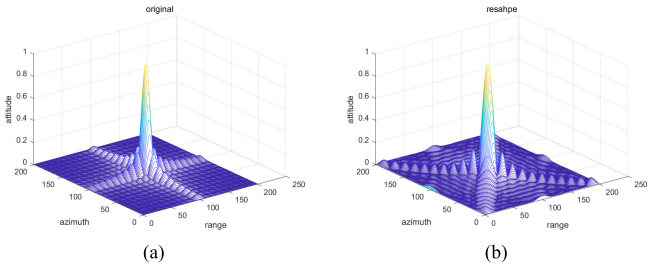


Fig. 7. Sidelobe directions before and after spectral reshaping. (a) Before. (b) After.

spectral reshaping are, respectively,

$$f(x, y) = \int\int F(x, y) e^{jk_x x} e^{jk_y y} d_{k_x} d_{k_y} \tag{8}$$

$$f'(x, y) = \int\int F'(x, y) e^{jk_x x} e^{jk_y y} d_{k_x} d_{k_y}. \tag{9}$$

After normalization, $f(x, y)$ and $f'(x, y)$ yield $f_1(x, y)$ and $f_2(x, y)$, respectively. The distributions of their sidelobes are shown in Fig. 7.

Fig. 7(a) shows the sidelobe direction of the original PSF, while Fig. 7(b) shows the changed sidelobe direction after the spectral reshaping operation. The imaging results of the original image data I_1 and the data after spectral reshaping I_2 are

$$I_1(x, y) = \sum_{m=1}^{M} \sigma_m f_1(x - x_m, y - y_m) \tag{10}$$

$$I_2(x, y) = \sum_{m=1}^{M} \sigma_m f_2(x - x_m, y - y_m) \tag{11}$$

where M represents the number of scattering points and σ_m denotes the scattering coefficient of the corresponding scattering point. x_m and y_m represent the positional coordinates

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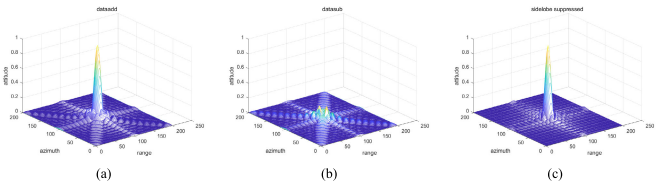


Fig. 8. PSF of (a) D_{add}, (b) D_{sub}, and (c) D_{final}.

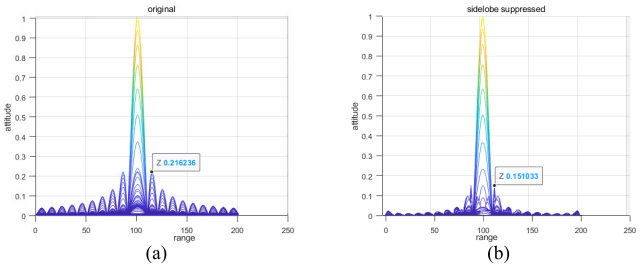


Fig. 9. Amplitude of the first sidelobe before and after the original SRSR method. (a) Before. (b) After.

of the scattering point. The mainlobe and sidelobes can be separated through simple logical operations. D_{add} represents the superimposed PSF of the mainlobe and sidelobes, while D_{sub} represents the PSF, where the mainlobe is separated. The PSF of D_{add} and D_{sub} are

$$D_{\text{add}} = \frac{1}{2}(|I_1(x, y) + I_2(x, y)|) \tag{12}$$

$$D_{\text{sub}} = \frac{1}{2}(|I_1(x, y) - I_2(x, y)|). \tag{13}$$

Here, D_{final} is the result after sidelobe suppression, obtained by

$$D_{\text{final}} = |D_{\text{add}} - D_{\text{sub}}|. \tag{14}$$

The PSF of D_{add}, D_{sub}, and D_{final} are, respectively, as shown in Fig. 8. With the original SRSR method, the majority of the sidelobes can be effectively suppressed, while the width of the mainlobe remains almost unchanged.

However, this method suffers from incomplete sidelobe elimination and blurred surroundings of strong scattering points. In the original SRSR method, it is common to reshape the spatial–spectral support area into a rhombus or parallelogram shape. This helps maximize the angle of sidelobe displacement, facilitating the separation of the mainlobe and sidelobes. Experimental results show that most of the sidelobes far from the scattering point center are suppressed. However, the sidelobe closest to the mainlobe has the shortest distance of displacement. After the shift, the resulting sidelobe still overlaps with part of the original sidelobe, indicating incomplete elimination of the first sidelobe. This can be seen from the comparison between Fig. 9(a) and (b), where the amplitude of the first sidelobe in the image processed by the original SRSR method remains relatively high.

After altering the sidelobe direction with the original SRSR method, incomplete suppression of the first sidelobe leads to the emergence of sidelobes in the nonorthogonal direction. The redundant nonorthogonal sidelobes blur the surroundings of strong scattering points, significantly reducing the quality of the SAR image. Fig. 10 shows the optical and SAR images

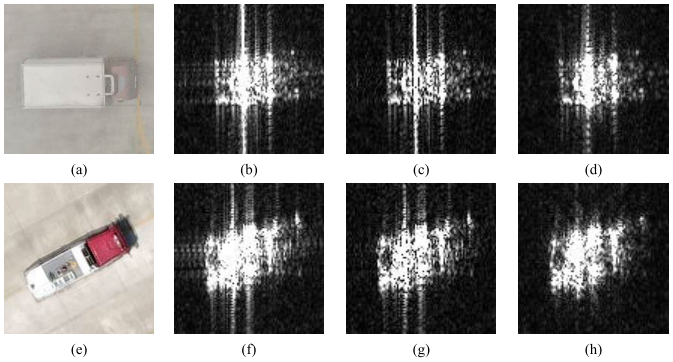


Fig. 10. Sidelobe suppression results of different methods for the two vehicles. (a)–(d) Optical image, original SAR image, SVA result, and the original SRSR result of vehicle 1, respectively. (e)–(f) Optical image, original SAR image, SVA result, and the SRSR result of vehicle 2, respectively.

of two vehicle targets. Fig. 10(a)–(d) represents the optical image, SAR image, SVA result, and the original SRSR result of vehicle 1, respectively. Fig. 10(e)–(h) represents the optical image, SAR image, SVA result, and the original SRSR result of vehicle 2, respectively. According to the basic principle of the SVA series algorithms, no additional sidelobes will be generated at strong scattering points. Hence, the SVA result can serve as a reference to examine the distribution of sidelobes around the scattering points after applying the original SRSR method. In Fig. 10(c), the strong scattering area of vehicle 1 becomes clear and visible with the SVA method, which was originally covered by sidelobes. However, in Fig. 10(d), the scattering structure in the same area becomes very blurry. The comparison of the suppression results of vehicle 2 in Fig. 10(g) and (h) further illustrates the shortcoming of the original SRSR method. Although the original SRSR method performs better than the SVA in terms of sidelobe suppression, it leads to blurs around strong scattering points, resulting in even worse visual effects compared with the unprocessed image. The SVA significantly outperforms the original SRSR method in preserving the clarity of scattering structures.

III. Proposed Method

To address the drawbacks of the original SRSR method in practical applications, this article proposes an improved SRSR method based on feature extraction. By utilizing the difference between the complex data of strong and weak scattering areas, a feature image containing only strong scattering points and sidelobes can be extracted. The modulus selection before subtraction further optimizes the feature image and reduces the energy loss of the mainlobe. The flowchart of our method is shown in Fig. 11.

The original SRSR method only performs logical operations on the SAR images before and after spectral reshaping to obtain the sidelobe suppression result. It does not consider the influence of the first sidelobe which is incompletely eliminated, resulting in poor sidelobe suppression performance. Unlike the original SRSR method, our method first utilizes the statistical differences in complex data of SAR images before and after spectral reshaping to construct two different sidelobe superposition matrices, $D3$ and $D4$. The scattering points in weak scattering areas of SAR images have almost

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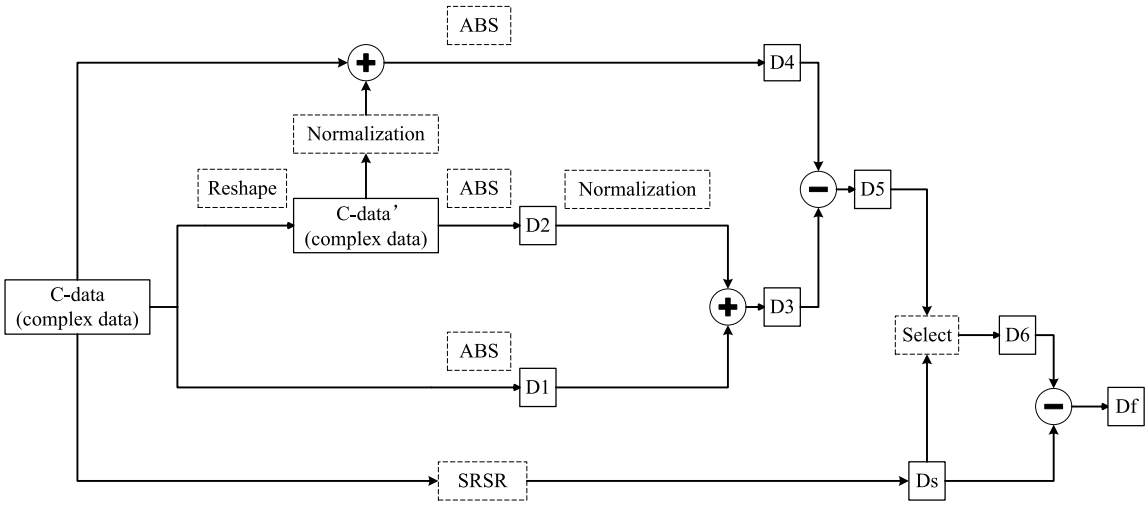


Fig. 11. Flowchart of the proposed method.

no sidelobes. Since the spectral reshaping operation mainly changes the sidelobe directions of strong scattering points, it has little effect on these scattering points without sidelobes. Therefore, in $D3$ and $D4$ matrices, weak scattering areas have small numerical differences, while strong scattering points and their sidelobe areas have large numerical differences. By subtracting $D4$ from $D3$, the modulus of the data in the weak scattering areas becomes 0. In the strong scattering areas, due to the significant differences in the absolute value of the data, it is feasible to extract data that includes both orthogonal and nonorthogonal sidelobes. This results in a feature matrix, $D5$, containing partial energy of the strong scattering points and the superimposed sidelobes. Then, data selection is performed on the matrix $D5$ and the result Ds of the original SRSR method. Specific selection rules are used to obtain matrix $D6$. The final result is obtained by subtracting $D6$ from Ds. The specific steps of the proposed method are as follows.

According to the theory of the original SRSR method, the spectral support area is reshaped and Fourier inverse transformed to obtain the complex data C-data$'$. C-data$'$ can be compute as follows:

$$C\text{-data}' = \sum (I_{\mathrm{SR}} + j \times Q_{\mathrm{SR}}). \tag{15}$$

Here, I_{SR} and Q_{SR} represent the real and imaginary parts of the complex data C-data$'$, respectively.

The absolute value of the original complex data is taken to obtain matrix $D1$, and the maximum value $\max_{D1}$ of the matrix $D1$ is taken as the normalization standard. The same operation is performed on C-data$'$ to obtain matrix $D2$ and the maximum value $\max_{D2}$. Matrix $D2$ is normalized and added to matrix $D1$ to obtain matrix $D3$. Matrix $D3$ is given by

$$D3 = D1 + \frac{D2 \times \max_{D1}}{\max_{D2}}. \tag{16}$$

Unlike matrix $D3$, matrix $D4$ is obtained by utilizing the difference between the complex data before and after spectral reshaping. The first step is to normalize I_{SR} and Q_{SR} with the previously obtained $\max_{D1}$ and $\max_{D2}$. Then, I_{SR} and Q_{SR} are added to the real part I_{ori} and imaginary part Q_{ori} of the original complex data, respectively, and then taking the absolute value to obtain matrix $D4$. Matrix $D4$ is given by

$$D4 = \left| \sum \left(I_{\mathrm{ori}} + \frac{I_{\mathrm{SR}} \times \max_{D1}}{\max_{D2}} \right) + \sum j \times \left(Q_{\mathrm{ori}} + \frac{Q_{\mathrm{SR}} \times \max_{D1}}{\max_{D2}} \right) \right|. \tag{17}$$

It can be deduced from the formula $|A| + |B| \geq |A + B|$ that

$$D3(i, j) \geq D4(i.j). \tag{18}$$

The purpose of constructing matrices $D3$ and $D4$ is to separate the orthogonal and nonorthogonal sidelobes. In SAR images, weak scattering areas such as grassland and concrete floors have almost no sidelobes. Since there are no sidelobes in weak scattering areas, the spectral reshaping operation has little influence on the magnitude and sign of the complex data in these areas. Therefore, for the complex data of weak scattering areas, whether taking the absolute value first and then adding or adding first and then taking the absolute value, the resulting magnitudes are almost equal. However, in strong scattering areas, especially at strong scattering points with cross-shaped sidelobes, after spectral reshaping operation, the sidelobe directions are drastically changed. As a result, the magnitude and sign of the complex data in both orthogonal and nonorthogonal directions undergo significant changes. In this case, for the complex data of strong scattering areas, the value obtained by taking the absolute value first and then adding will be greater than or equal to the value obtained by adding first and then taking the absolute value. Matrix $D5$ is precisely obtained based on this numerical difference. Matrix $D5$ is given by

$$D5 = D3 - D4. \tag{19}$$

In matrix $D5$, the magnitude of weak scattering areas will be close to 0. On the other hand, strong scattering areas will preserve both orthogonal and nonorthogonal sidelobes.

Fig. 12(a) shows an HH-polarized image collected by an airborne SAR with a ground resolution of 1.5 m. There are four

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Fig. 12. (a) SAR and (b) optical images of the same area.

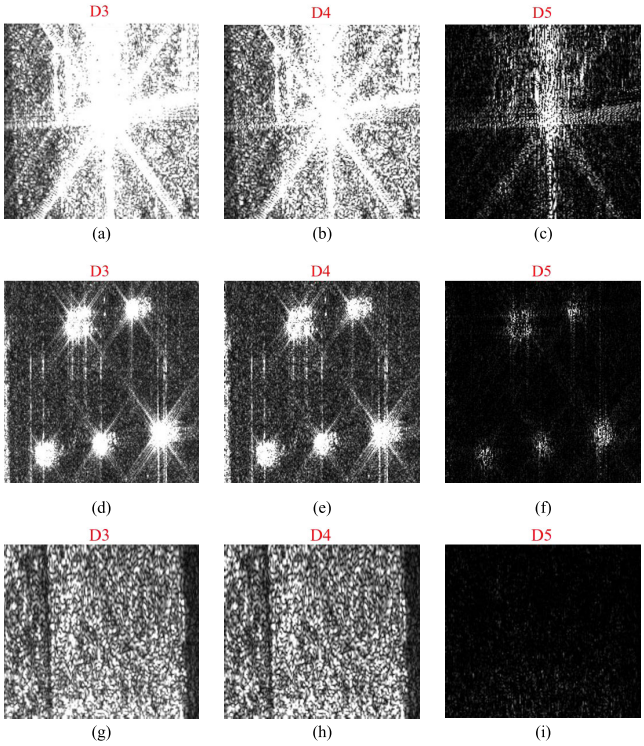


Fig. 13. Closed-view images of three regions in different matrices. (a)–(c) $D3$–$D5$ matrices for Region I. (d)–(f) $D3$–$D5$ matrices for Region II. (g)–(i) $D3$–$D5$ matrices for Region III.

types of targets in this SAR image: grassland, concrete floor, vehicles, and factories. Fig. 12(b) shows the corresponding optical image.

The optical image indicates that in Region I, the presence of the huge cross-shaped sidelobe is caused by the strong scattering structure of the rooftop of the factory. In Region II, there are vehicles and corner reflectors with sidelobes of different intensities. Region III represents a weak scattering area (grassland).

Fig. 13(a)–(c) depicts the imaging results of $D3$, $D4$, and $D5$ matrices for Region I, respectively. Fig. 13(d)–(f) shows the imaging results of $D3$–$D5$ matrices for Region II, respectively. Fig. 13(g)–(i) illustrates the imaging results of $D3$–$D5$ matrices for Region III, respectively.

It is clear that the strong scattering areas in Fig. 13(a) and (d) are brighter than the same areas in Fig. 13(b) and (e), respectively. Hence, the absolute value of the matrix $D3$ in the strong scattering area is greater than that of the matrix $D4$. By comparing Fig. 13(g)–(i), it can be observed that in the weak scattering area, there is little difference in the absolute value between matrix $D3$ and matrix $D4$.

Fig. 13(c) shows the result of subtracting the magnitude of (b) from (a). In Fig. 13(c), strong scatterers and their surrounding orthogonal and nonorthogonal sidelobes are retained, while the weak scattering regions appear as dark areas in the image, indicating that the pixel values in these regions are almost 0. Similarly, in Fig. 13(f), all five vehicle targets retain orthogonal and nonorthogonal sidelobes, and the weak scattering regions also appear as dark areas in the image. Fig. 13(c) and (f) shows that the matrix $D5$ preserves part of the energy of the strong scattering points, while also preserving the orthogonal and nonorthogonal sidelobes around them. Fig. 13(i) further shows that in the weak scattering areas, there is little scattering information preserved by the matrix $D5$.

Fig. 13 shows that the obtained matrix $D5$ contains the superimposed sidelobes of strong scattering points in both orthogonal and nonorthogonal directions. Assuming that Ds is the result of the original SRSR method. Then, subtracting $D5$ from Ds can significantly reduce the sidelobes generated by the original SRSR method. However, besides containing orthogonal and nonorthogonal sidelobes, matrix $D5$ unavoidably contains partial energy of the strong scattering points. If $D5$ is directly subtracted from Ds, it may result in energy loss of the mainlobe. Therefore, modulus selection is performed on Ds and $D5$. If the value of a certain pixel $D5(i, j)$ is smaller than the corresponding value $Ds(i, j)$, the value of that pixel in $D6$ is set to $D5(i, j)$. If the $Ds(i, j)$ is smaller than $D5(i, j)$, the value $D6(i, j)$ is set to 0. Matrix $D6$ is constructed by

$$D6(i,j) = \begin{cases} D5(i,j), & D5(i,j) < Ds(i,j) \\ 0, & D5(i,j) \geq Ds(i,j). \end{cases} \tag{20}$$

The final sidelobe suppression result Df is obtained by

$$Df = Ds - D6. \tag{21}$$

Through the modulus selection, the mainlobes with low intensity in Ds are preserved, avoiding causing energy loss of the mainlobes due to the subtraction of $D6$ and Ds. Simultaneously, $D5$ retains the sidelobes in both orthogonal and nonorthogonal directions. After the modulus selection, the modulus at the sidelobe positions in $D6$ will be less than or equal to the modulus at the corresponding sidelobe positions in Ds. After subtracting $D6$ from Ds, even if the sidelobes in both orthogonal and nonorthogonal directions cannot be completely eliminated, a significant reduction of sidelobes can still be achieved, effectively addressing the drawbacks of the original SRSR method.

IV. Experimental Results

A. Experiments on Airborne SAR Images

Fig. 14(a) presents an HH-polarized image collected by a Ku-band airborne SAR, with a resolution of 1.5 m and an image size of 1180×1460. The image was captured at an airport in Hunan Province, China. The optical image of the same area is shown in Fig. 14(b). Fig. 15 shows the sidelobe

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Fig. 14. (a) Optical and (b) SAR images.

suppression results with the MSVA, RSVA, the original SRSR method, and the proposed method.

To better compare the sidelobe suppression performance of different methods, three regions are selected for magnification in Fig. 14(a). The vehicle target in Region I has strong sidelobes, and the sidelobes at different structures overlap with each other, making it even more difficult to suppress. Region II exhibits a rooftop of a factory, which is a strong scattering area, where multiple sidelobes overlap to form a large cross-shaped sidelobe, seriously blurring the surroundings. In Region III, there is a metal box isolated in a grassy area, which can be approximated as a point target. These three regions have different scattering structures, which generate different types of sidelobes. By observing the sidelobe suppression results of these three regions, it is possible to comprehensively evaluate the performance of different sidelobe suppression methods in dealing with various types of sidelobes. Figs. 16–18, respectively, show the sidelobe suppression performance in Regions I–III.

In Region I, the strong scattering points of the vehicle target have cross-shaped sidelobes of varying intensities. In the case of multiple cross-shaped sidelobes gathering or even overlapping, the sidelobe suppression capability of the MSVA and RSVA significantly decreases. Both MSVA and RSVA can only suppress part of the sidelobes in the range direction, while almost no sidelobe suppression is performed in the azimuth direction, as shown in Fig. 16(a) and (b). Compared with the MSVA and RSVA, the original SRSR method has a better sidelobe suppression performance in both the azimuth and range directions. However, the original SRSR method obviously blurs the surroundings of the strong scattering points of the vehicle. Compared with the sidelobe suppression result with the MSVA and RSVA, the structures of the vehicle are less clear, as shown in Fig. 16(c). Fig. 16(d) shows the image with the proposed method. The sidelobe suppression performance in both the azimuth and range directions of the vehicle target is superior to that of the MSVA, RSVA, and the original SRSR method. Compared with Fig. 16(c), the phenomenon of blurred surroundings of the strong scattering points is significantly reduced, and the structures of the vehicle are clearly visible.

In Region II, the rooftop of the factory has a large number of strong scattering structures, which appear as multiple cross-shaped sidelobes superimposed on each other. In Fig. 17(a), the image with the MSVA shows poor performance on sidelobe suppression in the range direction. Fig. 17(b) shows the image with the RSVA. It is clear that RSVA performs better than MSVA in suppressing sidelobes in the range direction. However, RSVA causes significant energy loss of the mainlobe, leading to a notable decrease in the scattering intensity at the edges of the factory rooftop. Fig. 17(c) shows the image with the original SRSR method. The center of the strong scattering point in the image is seriously blurred by nonorthogonal sidelobes. Fig. 17(d) shows the image with the proposed method. The sidelobe suppression performance at the strong scattering points is significantly better than that of the MSVA and original SRSR methods. Although the proposed method shows slightly weaker sidelobe suppression performance in the azimuth direction compared with RSVA, it does not lead to the energy loss of the mainlobe. At the same time, there is no phenomenon of blurred surroundings caused by nonorthogonal sidelobes.

In Region III, the main target is an isolated metal box in the grassland. As shown in Fig. 18(a), with the MSVA, the sidelobes in the azimuth direction are significantly suppressed, but there are still remaining sidelobes in the range direction. Fig. 18(b) shows the result with RSVA. Compared with the MSVA, it performs better in the suppression of sidelobes in the range direction. However, the RSVA suppresses the mainlobe of the metal box, making it difficult to distinguish the metal box from the sidelobes in the range direction. Compared with the MSVA and RSVA, the original SRSR method better suppresses the sidelobes and preserves the mainlobe of the metal box. However, the incompletely eliminated sidelobes still cause blurs around the metal box, as shown in Fig. 18(c). Fig. 18(d) exhibits that our method further suppresses the sidelobes in both the range and azimuth directions. Compared with the original SRSR method, the phenomenon of blurred surroundings of the metal box is also greatly reduced, making the position and size information of the object more specific.

Fig. 19 presents the profile comparison of sidelobe suppression results of the isolated target in Region IV in both the range and azimuth directions. In terms of the range direction, the proposed method exhibits the smallest mainlobe width while maintaining consistent amplitude with the other three methods. In the azimuth direction, the proposed method still maintains the smallest mainlobe width, whereas MSVA and RSVA incorrectly determine the position of the mainlobe.

In order to quantitatively analyze the effectiveness of different sidelobe suppression methods, the peak sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR) metrics are used to evaluate the sidelobe suppression performance. The PSLR represents the ratio of the peak intensity of the strongest sidelobe I_s to the peak intensity of the mainlobe I_m. PSLR is often used to measure the ability to detect weak targets around strong targets. A smaller PSLR indicates that the energy of the targets is more concentrated, making them more prominent in SAR images. The PSLR is given by

$$\mathrm{PSLR} = 20 \cdot \lg \frac{I_s}{I_m}. \tag{22}$$

The ISLR is defined as the ratio of the sum of energy from all sidelobes in the impulse response function to the energy of the mainlobe. The ISLR metric focuses more on describing the importance of sidelobes relative to the mainlobe. The smaller

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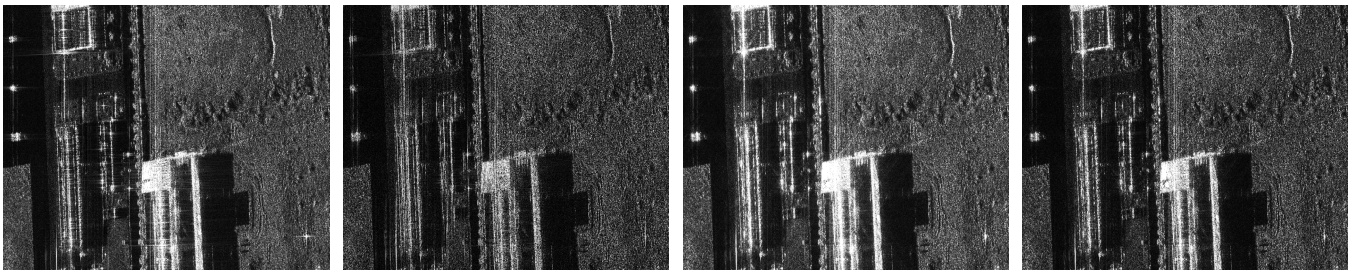


Fig. 15. Sidelobe suppression results of Ku-band SAR image with (a) MSVA, (b) RSVA, (c) original SRSR method, and (d) proposed method.

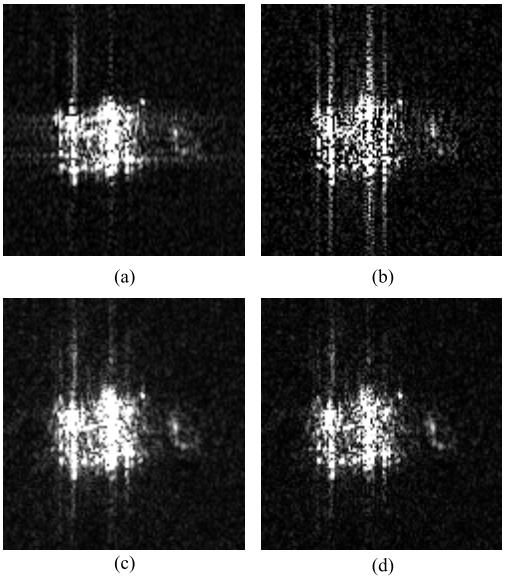


Fig. 16. Sidelobe suppression results of Region I with (a) MSVA, (b) RSVA, (c) original SRSR method, and (d) proposed method.

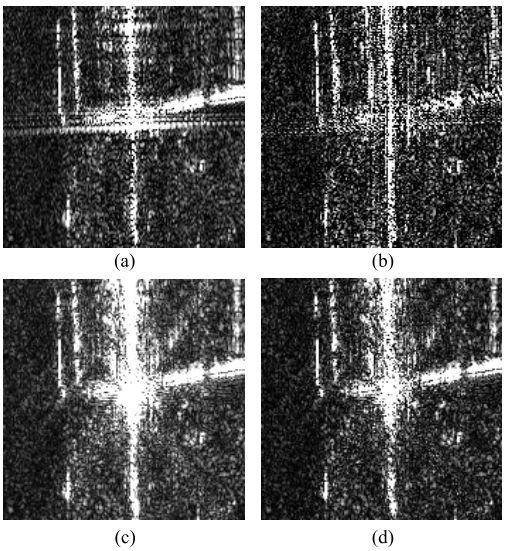


Fig. 17. Sidelobe suppression results of Region II with (a) MSVA, (b) RSVA, (c) original SRSR method, and (d) proposed method.

the ISLR, the stronger the ability to detect dark targets near bright targets. The ISLR is given by

$$\mathrm{ISLR} = 10 \cdot \lg \frac{P_{\mathrm{total}} - P_{\mathrm{main}}}{P_{\mathrm{main}}} \tag{23}$$

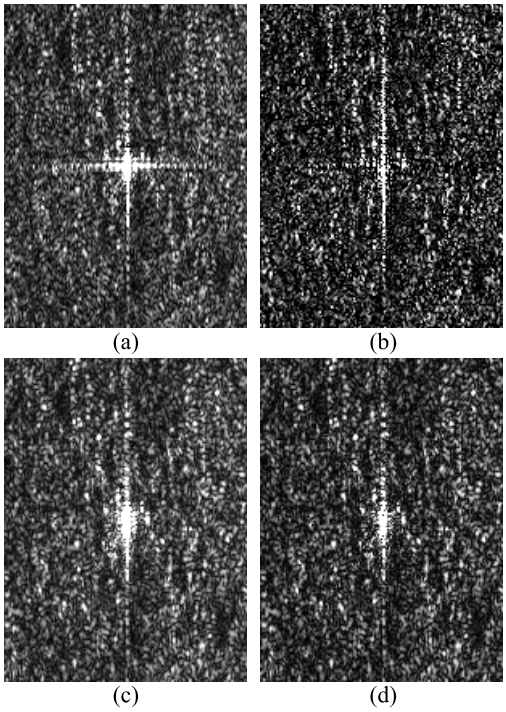


Fig. 18. Sidelobe suppression results of Region III with (a) MSVA, (b) RSVA, (c) original SRSR method, and (d) proposed method.

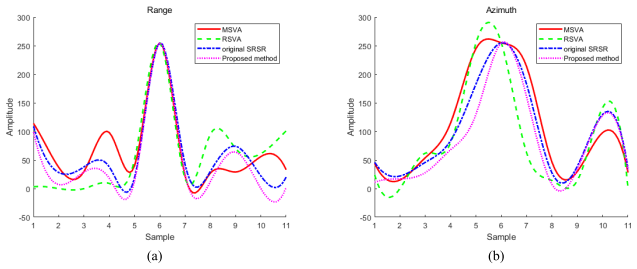


Fig. 19. Sidelobe suppression profile comparison of the isolated target in Region IV in terms of range and azimuth directions. (a) Sidelobe suppression performance of four methods in the range direction. (b) Sidelobe suppression performance of four methods in the azimuth direction.

where P_{total} represents the power in total and P_{main} represents the power of the mainlobe.

In order to evaluate the speed of image processing of the four methods, the processing time is introduced. In addition, the image entropy [33] is introduced to evaluate the quality of the four images. The smaller the image entropy, the lower the

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TABLE I
PSLR AND ISLR COMPARISON OF THE KU-BAND SAR DATA

	PSLR in azimuth(dB)	PSLR in range(dB)	ISLR in azimuth(dB)	ISLR in range(dB)
Original image	-7.4656	-28.6568	-9.8471	-26.2441
MSVA	-8.6093	-29.1759	-12.5730	-26.6830
RSVA	-4.5305	-28.7157	-6.8372	-29.2036
Original SRSR	-4.8049	-26.4394	-7.4561	-24.5471
Proposed method	-15.2836	-39.2543	-18.9920	-38.7975

TABLE II
IMAGE ENTROPY AND PROCESSING TIME COMPARISON OF KU-BAND SAR DATA

	Entropy(bit/pixel)	Processing time(s)
Original image	7.3931	-
MSVA	7.2808	3.5
RSVA	7.2540	5.80
Original SRSR	7.2201	0.77
Proposed method	7.0330	0.84

clutter interference, the clearer the image, and the higher the image quality. The image entropy is defined as follows:

$$H(x) = -\sum_{i=1}^{n=1} P(a_i) \times \log_2 P(a_i) \tag{24}$$

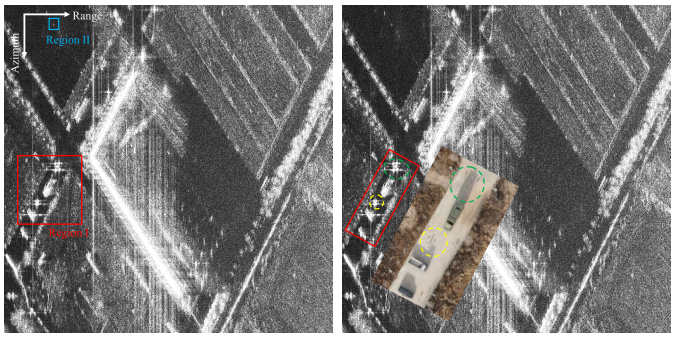
where $P(a_i)$ represents the probability of a certain pixel value occurring.

Table I shows the comparison of the PSLR and ISLR in both azimuth and range directions of the five images. Table I indicates that the proposed method can suppress the PSLR and ISLR in the range direction to −39 and −38 dB, respectively, which is significantly better than the MSVA, RSVA, and the original SRSR method. The sidelobe suppression performance in the azimuth direction is also significantly improved compared with the MSVA, RSVA, and original SRSR method. Table II shows the image entropy and processing time of the four methods. The results show that the Ku-band SAR image with the proposed method has the smallest image entropy. It means that the proposed method achieves more obvious sidelobe and clutter interference removal than other methods. In terms of processing speed, the proposed method also significantly outperforms MSVA and RSVA, demonstrating the advantage of spectral reshaping in sidelobe suppression.

To further verify the robustness of the proposed method, the above three methods are applied to an X-band SAR image. As shown in Fig. 20(a), the size of the image is 1800 × 1800 with a resolution of 1.5 m. Fig. 20(b) shows the correspondence between SAR and optical images of ground targets.

The green dashed circle contains five corner reflectors with a center distance of 2 m, and the yellow dashed circle contains five corner reflectors with a center distance of 1 m. Other targets are vehicles of different sizes.

Fig. 21 shows the sidelobe suppression results with MSVA, RSVA, the original SRSR method, and the proposed method. To further observe the sidelobe suppression performance on ground targets, Region I containing vehicles and corner reflectors in Fig. 20(a) is selected for magnification comparison. The results are shown in Fig. 22.



(a) (b)

Fig. 20. SAR image of the area and optical image of ground targets. (a) SAR image. (b) Correspondence between SAR and optical images of ground targets.

In the original SAR image, both corner reflector arrays have strong sidelobes, causing any two adjacent corner reflectors to be covered by each other's sidelobes. This phenomenon is particularly evident in the corner reflector array with the center distance of 1 m, where the boundaries of each corner reflector are very blurry. Fig. 22(a) shows that with the MSVA, the sidelobes at the corner reflectors and vehicles are suppressed to some extent. However, compared with the original SRSR method, the MSVA shows poor sidelobe suppression performance on processing the corner reflector array with the center distance of 2 m, as shown in Fig. 22(c). With the MSVA, the boundary of the corner reflectors cannot be clearly distinguished, and the sidelobe suppression performance in the azimuth and range directions is inferior to that of the original SRSR method. Fig. 22(b) shows the image with RSVA. Although RSVA outperforms MSVA in sidelobe suppression, RSVA results in significant energy loss of the mainlobe, which can be seen in the scattering intensity of the vehicle targets and corner reflectors. Fig. 22(d) shows the image with the proposed method. Compared with the MSVA, RSVA, and the original SRSR method, the sidelobe suppression performance in the corner reflector areas is much more significant. The relative positions and boundaries of the five corner reflectors are clearly visible. The result further indicates that our method not only performs well on the sidelobe suppression of corner reflectors but also on vehicle targets. Sidelobes in the azimuth and range directions are efficiently suppressed, while avoiding causing blurs around the strong scattering points. Structures of the vehicle with the proposed method are much clearer.

Fig. 23 presents the profile comparison of sidelobe suppression results of the strong scattering point in Region II in both the range and azimuth directions. In the range direction, both MSVA and RSVA exhibit varying degrees of energy loss of the mainlobe. The proposed method demonstrates the smallest mainlobe width in the range direction while preserving the mainlobe energy. In the azimuth direction, the proposed method still maintains the smallest mainlobe width, while MSVA and RSVA similarly suffer significant energy loss of the mainlobe.

Tables III and IV present the same metrics of the five images before and after sidelobe suppression. Table III demonstrates that the proposed method outperforms the MSVA, and original SRSR method, especially in terms of PSLR in range and ISLR

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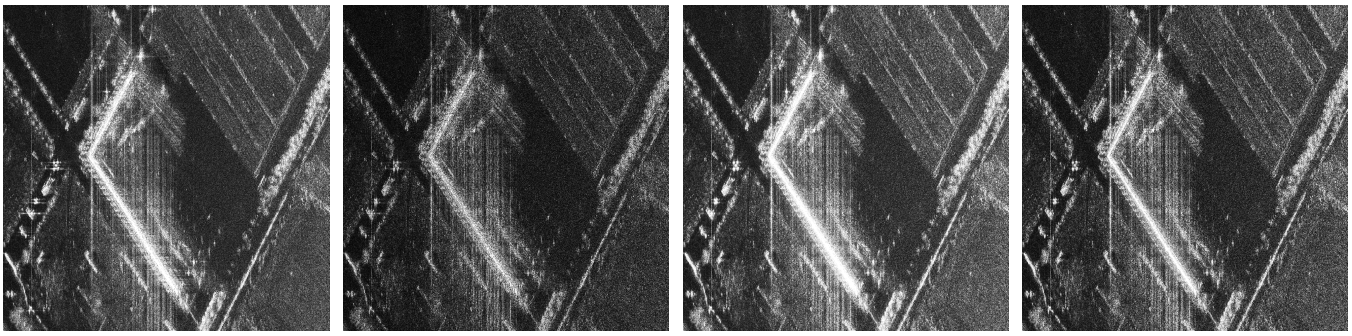


Fig. 21. Sidelobe suppression results of the X-band SAR image with (a) MSVA, (b) RSVA, (c) original SRSR method, and (d) proposed method.

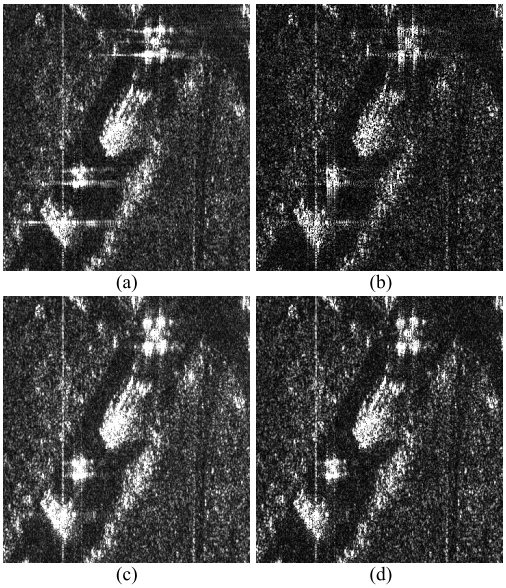


Fig. 22. Sidelobe suppression results of Region I with (a) MSVA, (b) RSVA, (c) original SRSR, and (d) proposed method.

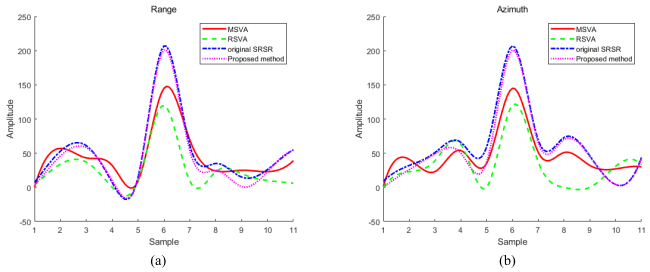


Fig. 23. Sidelobe suppression profile comparison of the isolated target in Region II in terms of range and azimuth directions. (a) Sidelobe suppression performance of four methods in the range direction. (b) Sidelobe suppression performance of four methods in the azimuth direction.

TABLE III

PSLR AND ISLR COMPARISON OF THE X-BAND SAR DATA

	PSLR in azimuth(dB)	PSLR in range(dB)	ISLR in azimuth(dB)	ISLR in range(dB)
Original image	-9.2710	-18.2450	-8.7686	-18.9704
MSVA	-9.2710	-18.2450	-11.0948	-19.1389
RSVA	-19.1457	-21.3202	-21.2063	-21.5951
Original SRSR	-14.2268	-23.1115	-14.6188	-24.1791
Proposed method	-19.1423	-29.5188	-18.4456	-31.9886

TABLE IV

IMAGE ENTROPY COMPARISON OF THE X-BAND SAR DATA

	Entropy(bit/pixel)	Processing time(s)
Original image	7.4746	-
MSVA	7.3951	5.74
RSVA	7.3373	13.42
Original SRSR	7.4317	1.21
Proposed method	7.3741	1.30

in range. Although RSVA slightly outperforms the proposed method in metrics PSLR in azimuth and ISLR in azimuth, RSVA results in energy loss of the mainlobe, leading to a significant decrease in image quality. In terms of preserving target information, it falls far behind the proposed method. Due to this particular drawback of RSVA, the image entropy of the image with RSVA is lower than that of the image with the proposed method, as shown in Table IV. It is obvious that our method runs much faster than MSVA and RSVA, as shown in Table IV.

B. Application on Target Detection

Target detection is one of the important applications of SAR image interpretation [34], [35], [36], [37]. In the images collected by spaceborne SAR, ship targets with strong sidelobes often appear, and the excessive sidelobes make it difficult to detect. At the same time, strong sidelobes also obscure surrounding weak scattering ships, making target detection even more challenging. Therefore, the effectiveness of sidelobe suppression in SAR image preprocessing significantly affects the accuracy of target detection.

Fig. 24 shows a SAR image slice of the Poyang Lake water area captured by the GF-3 satellite. The slice contains numerous ship targets, many of which have strong sidelobes. The results with the original SRSR method and the proposed method are shown in Figs. 25 and 26. With the original SRSR method, most sidelobes are suppressed, but there is still noticeable blurriness in certain strong scattering ships. However, with the proposed method, while further suppressing strong cross-shaped sidelobes, the phenomenon of blurriness of certain ships is significantly eliminated.

The target detection training model uses the open-source toolbox MMRotate in the OpenMMLab project, and the training network is oriented regions with convolutional neural networks (RCNN). The training set also uses the

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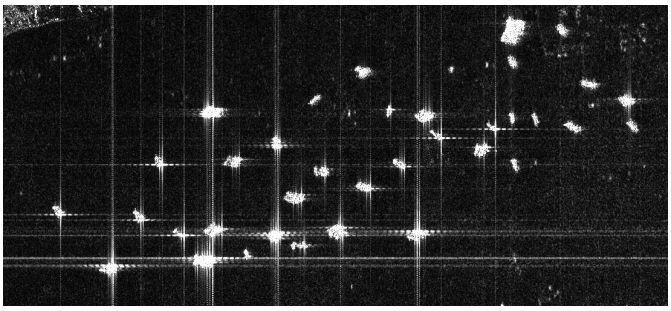


Fig. 24. Original SAR image.

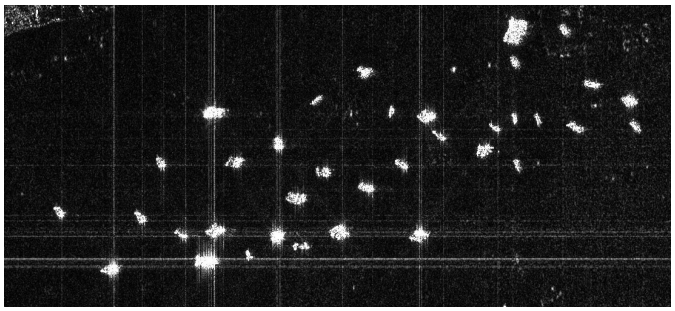


Fig. 25. Sidelobe suppression result with the original SRSR method.

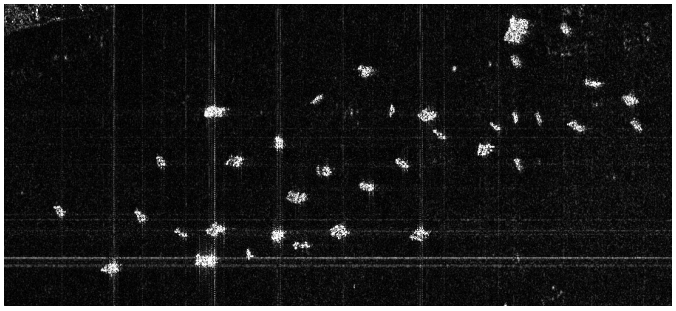


Fig. 26. Sidelobe suppression result with the proposed method.

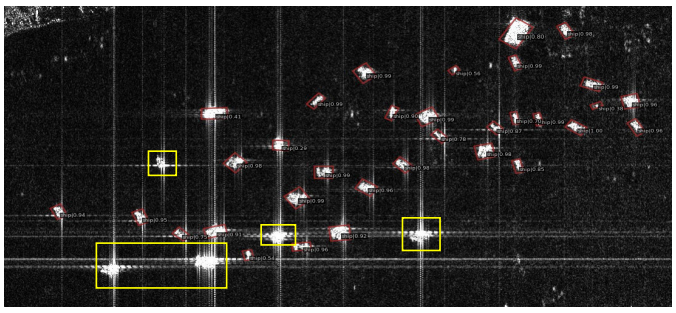


Fig. 27. Target detection result of the first original SAR image.

open source ship_detection_online and SSDD datasets. Figs. 24–26 are put as the test set, and the results of target detection are shown in Figs. 27–29.

Due to the influence of numerous sidelobes, there are cases of missed detection in the yellow box in Fig. 27. In addition, even for the detected ship targets, the presence of sidelobes distorts the detection boxes, making it difficult to accurately annotate the positions of the ships. Fig. 28 indicates that, based on the original SRSR method, some previously missed ship targets in Fig. 27 are detected due to sidelobe suppression, but there are still instances of missed detection and distortion

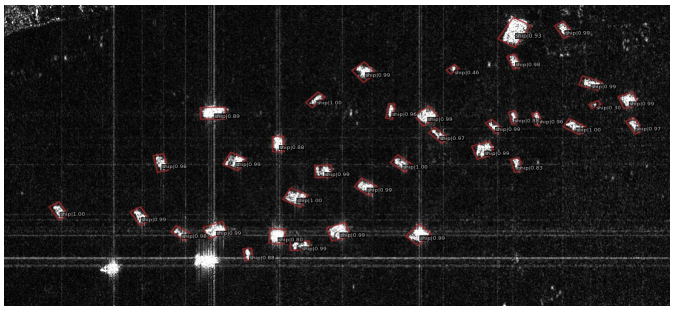


Fig. 28. Target detection result of the first SAR image based on the original SRSR method.

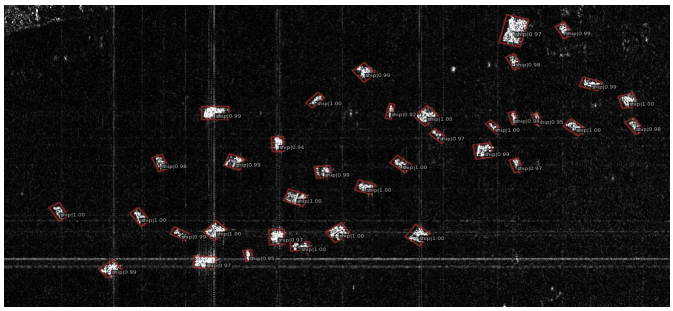


Fig. 29. Target detection result of the first SAR image based on the proposed method.

TABLE V
ACCURACY, PRECISION, AND TPR OF THREE DETECTION RESULTS

	Target detection result of the original image	Target detection result based on original SRSR	Target detection result based on the proposed method
Accuracy	0.769	0.871	0.923
Precision	0.906	0.917	0.944
TPR	0.829	0.943	0.971

of detection boxes. Fig. 29 shows the target detection result based on the proposed method. Due to the better sidelobe suppression performance of our method, more ship targets have been detected. The distortion of the detection boxes caused by sidelobes is also resolved. Table V uses accuracy, precision, and true positive rate (TPR) as evaluation metrics to further quantitatively analyze the above three target detection results.

The results in Table V demonstrate that the SAR image with the proposed method can achieve higher accuracy, precision, and TPR in target detection. This means that our method surpasses the original SRSR method in terms of sidelobe suppression performance, making it more suitable for SAR image preprocessing before target detection.

To further verify the robustness of the proposed method in improving the target detection rate, a new set of target detection experiments has been added. In this set of SAR images, all the ship targets are elongated and densely distributed, which is very different from the previous experiment. Fig. 30 shows the target detection results of the SAR image without sidelobe suppression. Figs. 31 and 32 show the target detection results based on the original SRSR method and the proposed method, respectively.

In Fig. 30, several ships cannot be accurately detected due to strong cross sidelobes. Fig. 31 shows the target detection result based on the original SRSR method. Since this method

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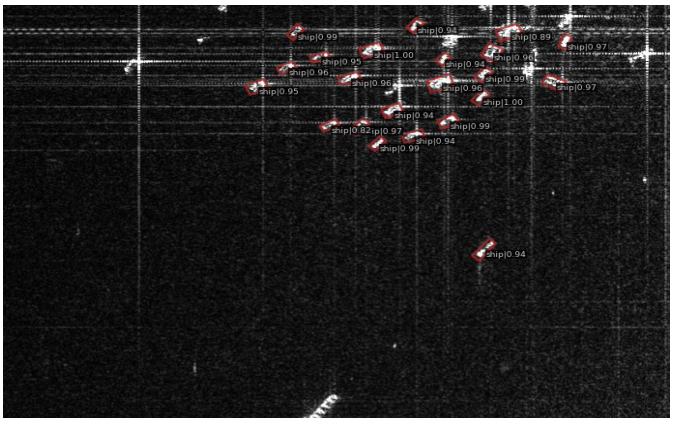


Fig. 30. Target detection result of the second original SAR image.

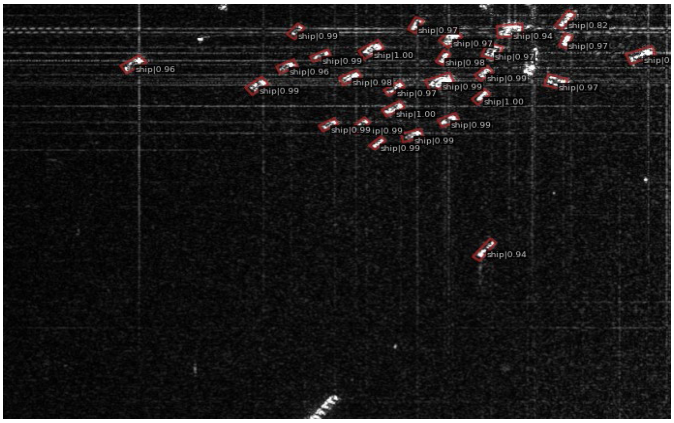


Fig. 31. Target detection result of the second SAR image based on the original SRSR method.

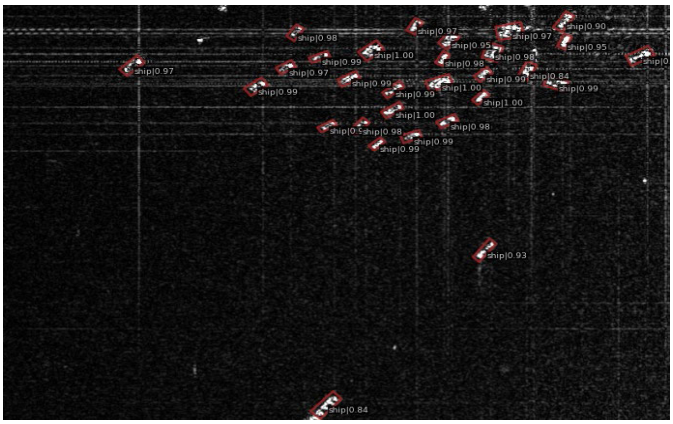


Fig. 32. Target detection result of the second SAR image based on the proposed method.

effectively suppresses the sidelobes, the targets with strong sidelobes in Fig. 30 can be detected. However, there are still cases of missed detection in the area where multiple ship targets are gathered. Fig. 32 shows the target detection result based on the proposed method. In Fig. 32, all ship targets are detected due to the better sidelobe suppression performance of our method.

The two experimental results show that for different types of ship targets, after preprocessing with the proposed method, the target detection rate can be significantly increased.

V. Conclusion

Based on the idea of extracting the sidelobe superposition feature, this article proposes an improved spectral reshaping sidelobe suppression method. By utilizing the statistical differences between the complex data in orthogonal and nonorthogonal sidelobe areas before and after spectral reshaping, the strong scattering points and their surrounding superposition sidelobes can be accurately extracted. Modulus selection is then performed on the extracted feature image before the final subtraction to avoid energy loss of the mainlobes. The proposed method solves the problem of incomplete suppression of nonorthogonal sidelobes in the original SRSR method, which would cause blurs around the scattering points. In addition, it better suppresses strong cross-shaped sidelobes. Comparative experiments on Ku-band and X-band airborne SAR images show that our method has significant advantages in suppressing sidelobes in the situation of multiple strong scattering points gathering. Compared with the original SRSR method, the proposed method maintains the clarity of the target scattering structures while suppressing sidelobes, significantly improving visual effects. Quantitative analysis experiments demonstrate that our method outperforms the MSVA, RSVA, and original SRSR method in terms of PSLR, ISLR, processing time, and image entropy evaluation metrics. The experiment on multiband SAR images further proves the robustness of our method. The experimental results on GF3 data show that the target detection result based on our method has a significant improvement in accuracy, precision, and TPR.

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